# Sentiment Analysis Tool for Spanish Tweets in the Ecuadorian Context

Ismael Utitiaj*
Department of Computer Science
Universidad Politécnica Salesiana
Quito, Ecuador
eutitiaj@est.ups.edu.ec

Paulina Morillo
IDEIAGEOCA Research Group
Universidad Politécnica Salesiana
Quito, Ecuador
pmorillo@ups.edu.ec

Diego Vallejo-Huanga [12]
[1] IDEIAGEOCA Research Group
[1] Universidad Politécnica Salesiana
[2] Department of Physics and Mathematics
[2] Universidad de las Américas
Quito, Ecuador
dvallejoh@ups.edu.ec
diego.vallejo.huanga@udla.edu.ec

## ABSTRACT

The huge amount of textual information that exists on social networks added by users through comments, has aroused a great interest in companies and research groups, which seek to use this information to identify trends and acceptance levels of brands, products, and services. A technique to know the level of acceptance or rejection of a particular topic, in an automated way, is the Sentiment Analysis. Some informatics tools incorporate this technique, however, there are few contributions to texts in Spanish. It is because of the difficulty of identifying different contexts, dialects, complex grammatical structures, and semantic language variances in each region. This article presents a web tool for the analysis of sentiments in texts written in Spanish that include Ecuadorian dialect or idioms. The tool was developed in R-Shiny with an approach lexicon-based. The tool allows the customization of the lexicons based on context and facilitates the automatic download of tweets according to search criteria such as the place, dates, and topic. To evaluate the effectiveness of the application, their result was compared with two commercial tools (Azure Text Analytics and IBM Watson NLU) and a manual score carried out by a group of people. The tests include the analysis of three corpora created from tweets. The results show the effectiveness of the tool to identify the sentiment polarity, especially in texts that include dialects, colloquial words, and negative expressions.

## CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; • **Information systems** → *Document topic models*; Open source software.

## KEYWORDS

R-Shiny, Azure Text Analytics, IBM Watson NLU, Twiter, Lexicon

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## 1 INTRODUCTION

Sentimental analysis is one of the most addressed tasks in the field of natural language processing (NLP), which seeks to identify subjective information in a word or sentence and determine the feeling (typically negative, positive, or neutral labels) that accompanies the text [20]. In recent years, the sentimental analysis has become very important, due to the high volume of textual data available on the Internet, especially on social networks. The opinion issued by users on several topics, such as politics, marketing, industry, services, etc., serves to measure the emotional state of people and, consequently, facilitates decision-making [3, 7]. In the case of the social network Twitter, there are many research found in the literature [8, 11, 12, 24], due to the popularity of the social network, and also, because it offers developers the possibility of consuming their data, through an API, this allows the download and analysis of tweets.

The models developed for sentiment analysis are based on two approaches. The first based on the use of lexicons and the second based on Machine Learning (ML) techniques [12, 17]. Both approaches are applied in [5] for polarity detection, through a model base on classifier ensembles and lexicon. This same work presents a summary, from 2005 to 2013, of the main contributions on sentiment analysis of tweets. In 2016, the work [20] presented a SentiCircles, a lexicon-based approach, which allows the detection of feelings both at the entity level and tweet level, taking into account co-occurrence patterns of words in different contexts in tweets. On the other hand, several research uses deep learning to determine offensive messages or determine the classification of feelings in tweets [1, 10, 15, 22, 25]. For instance,[6] presents the design a deep convolutional network for the sentiment analysis of short text, the model is tested on the Stanford Twitter Sentiment (STS) corpus, reaching an accuracy of 86.4%.

Although, there are several works that address sentiment analysis, on tweets written in English, contributions in the Spanish

*Both authors contributed equally to this research.

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language are still scarce. A recent contribution [4] presented Sentymental Analysis Model (SAM), a model that integrates various fields such as linguistics, artificial intelligence, and information retrieval. This work provides a basis for describing opinion classification systems in Spanish. In [19], the comparison of two models is shown, the first one based on Support Vector Machines (SVM) with linear kernel, representing the comments in a bag of words with their respective Term-Frequency Inverse-Document-Frequency (TF-IDF) and the second based on Convolutional Neural Networks (CNN), the latter achieved an accuracy of 82%. Another investigation carried out from 2017 to 2019 by the University of Malaga, presents a Sentiment Analysis tool called Lingmotif [14]. The solution is based on a lexicon of words and phrases. Each word in the lexicon is classified with positive and negative values to identify its tendency. The effectiveness of this tool lies in the application of context rules and an appropriate rating for each word or linguistic expression. Additional works [2], apply general concepts of data mining, to extract the opinion of Twitter users regarding specific topics.

At the same time, as the techniques have evolved in their effectiveness and precision, new tools for Sentiment Analysis have appeared, such as Natural Language Toolkit (NLTK) Sentiment Analyzer, TextBlob, Standford CoreNLP, RapidMiner, etc. As shown in [21], there are also several free access web tools and services that address the task of sentiment analysis.

In the case of sentiment analysis tools for Spanish texts, most of them are based on translation, that is, the sentences in Spanish are translated into English, before determining their polarity. This reflects major deficiencies regarding advances in the English language, where the techniques involved already work at the level of grammatical structures. Also, they address the problems of subjectivity and sentence context, using mathematical models more precise and focused on improving the calculation performance of the polarity of the feeling of each word. Besides, when analyzing the opinions of users on social networks, other aspects such as dialects, languages, the use of colloquial phrases, and the context of each region must be taken into account.

Supported by the suite of tools provided by the R platform, this work shows the development of a free access web tool, called Sentimental Analysis on Ecuadorian Tweets (SAET), which aims to incorporate idioms and Spanish words in the Ecuadorian context. The tool performs a personalized sentiment analysis of the text, calculates the polarity of each tweet, and creates interactive visualizations of the results.

## 2 MATERIALS AND METHODS

### 2.1 Development environment and libraries

The programming language used in the development of the web application was R v.4.0.2, under the R Studio v.1.0.143 development environment. R is an open-access software that facilitates data processing and graphical presentation in an agile and simple way. Besides, it has a specialized library for web applications called Shiny [16]. The applications developed with this library are based on the Client-Server architecture and are highly reactive and responsive, giving users effective and interactive feedback. The advantage of using this library lies in the fact that it is responsible for automatically generating the HTML5 / JavaScript / CSS necessary to build the Web application with parameters and dynamic variables that allow interacting with the data and seeing different results without resorting to complex programming. Additionally, it provides a ShinyApps.io host that allows up to five applications to be hosted for free. On the other hand, R has built-in several robust libraries that allow us to carry out Data Analysis and Sentiment Analysis. The most relevant package used to the sentimental analysis was *SentimentR* [18]. Our developed application is available from https://sentimentalupstool.shinyapps.io/SentimentAnalysisTrend/.

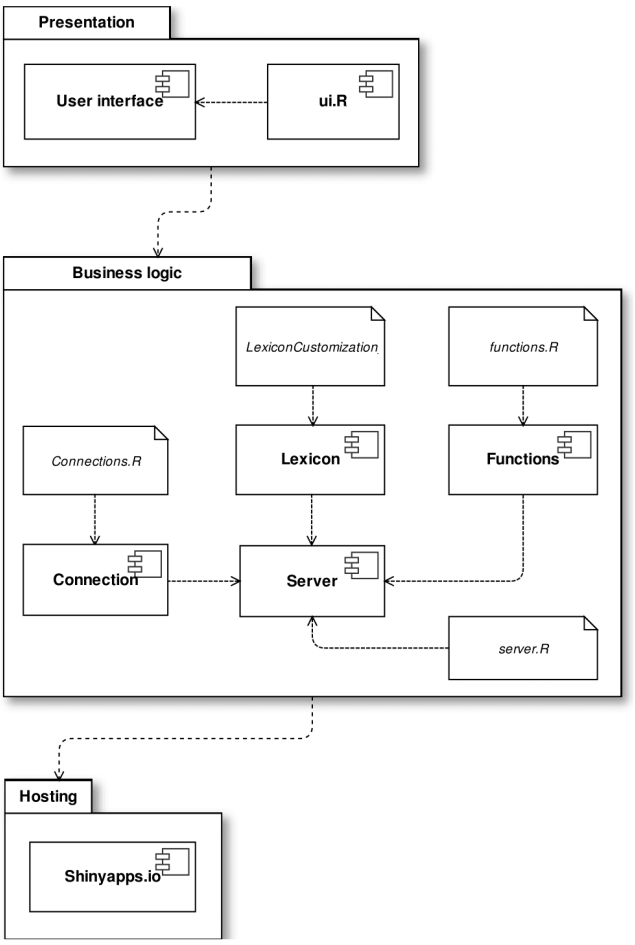



**Figure 1: Schema of SAET files**

### 2.2 Design and construction of the tool

Under the Client-Server architecture, Shiny requires the generation of at least two R files: the first for the user interface, named *ui.r*, and the second for the server, named *server.r*. In this way, the interface, including the screen elements, are declared in the *ui.r* file; and the server logic and dynamic content generated by screen interactions, are declared in the *server.r* file. Another alternative is to create a single file called *app.r* that contains both the interface and server elements. In this research, a solution with files separated by functionality and interface was used.

The SAET application contains five .r files, four of them take care of the application's functionality. Thus, the *LexiconCustomization_EC.r* file, has the declaration and implementation to load the lexicons, *functions.r* contains all the functions for creation and cleaning corpus, Document-Term Matrix (DTM) construction, polarity calculation, and visualizations of results. *Connection.r*, allows connection with Twitter, for searching and downloading tweet, and finally *server.r* contains all the server logic related to the components added in the interface of the user. On the other hand, *ui.r* uses a standard template called a shiny dashboard to control the design and appearance of the application such as navigation bars, dialog boxes, and dynamic navigation. The files described form the basic architecture of the application in Shiny. Figure 1 shows the structure of the files. All codes and files are available from https://github.com/iwiaCorp/SATrends/.

The interface application has four tabs. The first one defined as Local Data allows the analysis of sentiments of a corpus of tweets that are loaded locally through a file in CSV format. The second tab called Online Data allows download tweets directly from Twitter, using the Twitter API for R. The parameters required for the connection are *Consumer key, Consumer secret, Access token, Access secret*. These parameters were managed through a request sent to Twitter for approval and delivery of access credentials. With the *setup_twitter_oauth()* function the authentication and connection is performed. After establishing the connection, the tweets are obtained using the *searchTwitter()* function. In this function, the search word, a date range, and a geolocation radius were sent. The information of each tweet obtained from the Twitter Social Network are the text of tweet (*text*), the creation date in the format of DD/MM/YYYY (*created*), and the username (*screenName*). The third Lexicon Adjustment tab allows us to view the lexicon used by the tool and make modifications such as editing, removing, and adding new words and changing their polarity value. The results of the analysis are represented in interactive visualizations that allow the user to identify the trend of feelings. Furthermore, the polarity degree result of each tweet can be downloaded in a CSV file. The fourth tab corresponds to Help, in this section the loading procedure for local data and how Sentiment Analysis is performed is defined in a general way.

## 2.3 Sentiment Analysis Setup

The sentiment analysis process carried out by the tool requires three processes that are described in the diagram in Figure 2.

*2.3.1 Lexicons and Polarity Estimate.* In this phase, it is necessary to build two lexicons in Spanish. The first one, named as Polarity Lexicon (L1), is based on the English lexicon generated by the National Research Council Canada (NRC) [13] and translated automatically into Spanish, this dictionary shows two columns (word - polarity value) and it is stored in a CSV file. It contains 2514 words, of which 2041 are positives and 473 negatives. The polarity value is the one assigned by default by the NRC. This CSV file was manually adjusted and debugged, to remove duplicate and mistranslated words, and to add new colloquial words and phrases used in Ecuador. Thus, L1 contains a total of 3771 words, 2509 positive, and the rest negative.

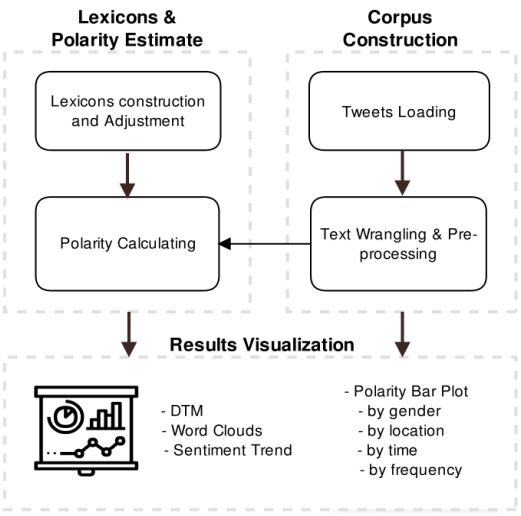



**Figure 2: Sentiment Analysis Setup**

The second lexicon, called Sentiment Valence Shifters (L2), aims to adjust the polarity to the most natural sentiment of the word or phrase and is made up of words such as negators (1), amplifiers (2), de-amplifiers (3), and adversative conjunctions (4). For example, in the phrases "I like" and "I like it a lot", it can be concluded that the latter denotes a greater intensity of positive polarity than the former. L2 contains 146 words, 37 of them negators, 78 amplifiers, 21 de-amplifiers, and 10 adversative conjunctions. The construction of this lexicon is based on dictionaries' automatic translation of the *Lexicon package* and has been refined so as not to add words that were already in the Polarity Lexicon.

Once the two dictionaries were built, the *Sentiment* function from the *SentimentR* package was used to calculate the polarity. The main parameters require are polarity lexicon (*polarity_dt*), sentiment valence shifters dictionary (*valence_shifters_dt*), and the text to be analyzed (*text.var*). The function supports a series of additional parameters to improve the polarity calculation, but some of them are applicable only for the English language. In our case, four additional parameters were used that can be applied to texts in Spanish.

- `amplifier-weight`: Value from 0 to 1 to apply to amplifiers/de-amplifiers. It will multiply the polarized terms by 1 plus this value (default 0.8).
- `adversative.weight`: The weight to give to the adversative conjunctions or contrasting conjunctions that annul the previous clause [9].
- `n.before`: The number of words (5) to consider as valence shifters before the polarized word.
- `n.after`: The number of words (3) to consider as valence shifters after the polarized word.

The polarity (Pol) calculation equation is described in the *SentimentR* package documentation within the *Sentiment* function. This function analyzed entire corpus and returns a polarity value for each tweet, theoretically defined in the interval $]-\infty, \infty[$. The polarity value depends on the length of the tweet and the number

of sentiment displacers found in the expression. This value was normalized by dividing it by the maximum polarity found in the corpus. The label of the tweet (negative, neutral, and positive) was determined by rules describe in Equation 1. These rules also apply to the other tools being compared Azure Text Analytics (ATA) and IBM Watson NLU (IBM-W).

$$label = \begin{cases} negative & if\ \ Pol < -0.05 \\ neutral & if\ \ -0.05 \leq Pol \leq 0.05 \\ positive & if\ \ Pol > 0.05 \end{cases} \quad (1)$$

*2.3.2 Corpus Construction.* The SAET tool allows the manual upload of tweets through a CSV file or the automatic download of tweets through the consumption of the API. Once the data is loaded, it is necessary to create a corpus, to have a compact and structured data set. The corpus is a collection of text, where the information of each tweet is stored. With the generated corpus and before performing an analysis, it is necessary to debug and clean the data. As a first step, all the text is converted to lowercase. Then the punctuation marks, numbers, stopwords, are removed in the text, except for a set of words that are important in the formation of sentences, these words are *no, tenemos, son, la, las, en, es, todo, ante*. As the last step, the characters related to web addresses, double spaces, emoticons are removed. The corpus is stored in a CSV flat file and incorporates, in addition to the initial information downloaded from Twiter, data such as a username, name, surname, gender [23], and country.

*2.3.3 Visualization Results.* Finally, the results of the sentiment analysis are shown through visualizations and reports. One of the reports presented by the tool is the document-term matrix, this matrix identifies the relevant words in the corpus, employing the frequency with which they appear in each text (tweet). The terms are located in rows and the document's id in the columns.

In addition, seven more visualizations are included, showing different perspectives on the results. The first is a diagram of the sentiment trend of the corpus, depending on the polarity of each tweet. The second is a bar chart that classifies the total of negatives, neutrals, and positives tweets. The third is a presentation of meaningful words using a word cloud. The fourth is a count of words classified by negative, positive, and a percentage of the 10 most used. The remaining three graphs visualize sentiments with smooth line graphs, horizontal bars, and dots, these applied to present information by day, gender, and city, respectively.

## 2.4 Tool Evaluation

The SAET tool was compared with the results of two commercial tools ATA and IBM-W. Every tool was evaluated in contrast to a manual assessment carried out by a group of people. Also, they were tested on three corpora, which were built from tweets on different topics. The search criteria and the average polarity of manual evaluation of each tweet are shown in Table 1.

*2.4.1 Comercial Sentiment Analysis Tools Setup.* The Azure Text Analytics tool requires a subscription account in Azure and to enable a Cognitive Service called Text Analytics. The input will be a CSV file, with three columns that correspond to the text identifier (ID), the text to be analyzed, and the language. After the CSV file is imported, the Power Query editor tool is used to add the connection steps to Azure Cognitive Services. The application returns a file with the ID of each tweet and its polarity value, the scale used goes from zero to one, where zero is the most negative value. To establish a homogeneous and comparable scale between the three tools, it was necessary to apply a normalization, which is shown in 2. Where P is the polarity value, 0.5 corresponding to the Azure Text Analytics scale midpoint value.

**Table 1: Search criteria for tweets**

| | Corpus 1 | Corpus 2 | Corpus 3 |
|---|---|---|---|
| Date | 01/07/2019 22/07/2019 | 08/10/2019 17/10/2019 | 16/10/2019 26/10/2019 |
| Place | Ecuador | Ecuador | Ecuador |
| Topic or User | Supermaxi Ecuador (@Supermax-iEcu) | Lenin Moreno Presidente Ecuador (@Lenin) | Paro nacional (#paro) |
| General Topic | supermarket brand. | political person. | economic and social situation of the country. |
| $\overline{Pol}$ | -0.0295 | -0.1596 | -0.0352 |

$$Pol = 2(P - 0.5) \quad (2)$$

In the case of IBM Watson NLU, a registered account in the IBM Cloud is required and to contract the Natural Language Understanding (NLU) service. A Python script can be used to connect to the Watson NLU service. The structure of the CSV file consists of three columns, the first to specify the text to be analyzed, the second to incorporate the creation date of the tweet, and the last to place the user's name. Similarly, the tool returns a polarity value, defined in the interval $[-1, 1]$.

*2.4.2 Manual Evaluation Methodology.* This process consisted of a manual review of each tweet by six different people. We had 60 collaborators at this stage, who were Ecuadorian, Spanish native speakers, with university education, and their ages ranged from 18 to 35 years old. Each corpus was divided into three parts of approximately 350 tweets. The nine groups of tweets were randomly distributed and independently reviewed. Each person had to rate each tweet in their assigned group on a scale of -1 to 1, and also assign a label Negative (N), Neutral (NE), or Positive (P), depending on the sentiment expressed in the text. The assessment results were consolidated into a single file and verified to avoid contradictions between the score and the label. To estimate the final polarity value (Pol), the average of the scores given by each evaluator was used.

*2.4.3 Evaluation Metrics.* Supervised Machine Learning metrics such as Accuracy (Acc), Precision (Pr) and Recall (Re), were used to measure the performance of the tools in the classification task. The labels obtained in the manual evaluation are considered as ground truth (real class). In addition, the average polarity of each corpus was determined.

**Table 2: Benchmark of performance evaluation results of sentiment analysis tools: SAET, Text Analytics, and IBM Watson.**

| | | Predicted Class | | | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Corpus 1 | | | | | | | | | Corpus 2 | | | | | | | | | Corpus3 | | | | | | | | | |
| | | SAET | | | Text Analytics | | | IBM Watson | | | SAET | | | Text Analytics | | | IBM Watson | | | SAET | | | Text Analytics | | | IBM Watson | | |
| | | N | NE | P | N | NE | P | N | NE | P | N | NE | P | N | NE | P | N | NE | P | N | NE | P | N | NE | P | N | NE | P |
| **Actual Classes** | N | **440** | 117 | 98 | **368** | 65 | 222 | **373** | 246 | 36 | **310** | 126 | 119 | **305** | 71 | 179 | **201** | 331 | 23 | **193** | 69 | 116 | **146** | 43 | 189 | **133** | 231 | 14 |
| | NE | 10 | **29** | 21 | 10 | **25** | 25 | 8 | **43** | 9 | 15 | **26** | 36 | 24 | **22** | 31 | 6 | **67** | 4 | 27 | **19** | 27 | 21 | **14** | 38 | 22 | **46** | 5 |
| | P | 20 | 42 | **116** | 21 | 16 | **141** | 13 | 70 | **95** | 54 | 56 | **154** | 42 | 32 | **190** | 29 | 163 | **72** | 69 | 43 | **187** | 61 | 33 | **205** | 43 | 220 | **36** |
| | Pr | 0.94 | 0.15 | 0.49 | 0.92 | 0.24 | 0.36 | 0.95 | 0.12 | 0.68 | 0.82 | 0.13 | 0.50 | 0.82 | 0.18 | 0.48 | 0.85 | 0.12 | 0.73 | 0.67 | 0.15 | 0.57 | 0.64 | 0.16 | 0.47 | 0.67 | 0.09 | 0.65 |
| | Re | 0.67 | 0.48 | 0.65 | 0.56 | 0.42 | 0.79 | 0.57 | 0.72 | 0.53 | 0.56 | 0.34 | 0.58 | 0.55 | 0.29 | 0.72 | 0.36 | 0.87 | 0.27 | 0.51 | 0.26 | 0.63 | 0.39 | 0.19 | 0.69 | 0.35 | 0.63 | 0.12 |
| | $\overline{Pr}$ | | 0.53 | | | 0.51 | | | 0.58 | | | 0.48 | | | 0.49 | | | 0.57 | | | 0.46 | | | 0.42 | | | 0.47 | |
| | $\overline{Re}$ | | 0.60 | | | 0.59 | | | 0.61 | | | 0.49 | | | 0.52 | | | 0.50 | | | 0.47 | | | 0.42 | | | 0.37 | |
| | Acc | | 0.66 | | | 0.60 | | | 0.57 | | | 0.55 | | | 0.58 | | | 0.38 | | | 0.53 | | | 0.49 | | | 0.29 | |
| | $\overline{Pol}$ | | -0.0006 | | | 0.0303 | | | -0.0947 | | | -0.0132 | | | 0.0076 | | | -0.1046 | | | -0.0007 | | | 0.03604 | | | -0.11278 | |

## 3 RESULTS AND DISCUSSION

The sentiment analysis results of the three corpora were summarized in Table 2. Although the polarity rating of the sentiment of each tweet is subjective, it could be verified, through the sentiment trend of the corpus (Figure 4), that the trend of manual polarity was quite similar to the trend shape of the other three tools. However, manual evaluation tends to be more negative.

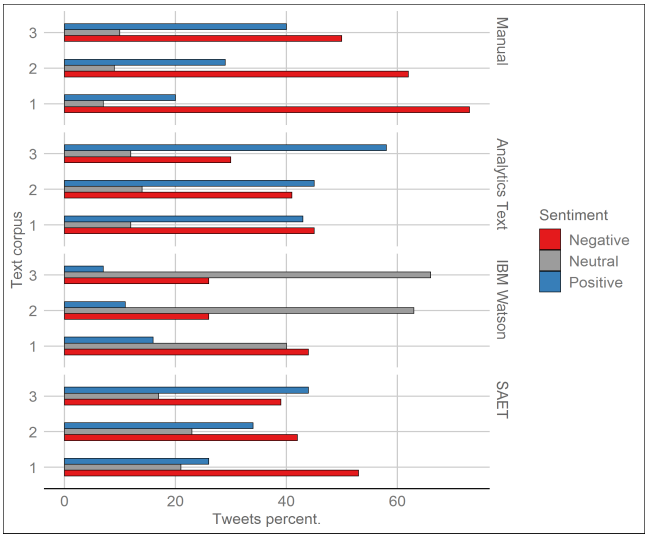



**Figure 3: Percentage of tweets (negative, neutral and positive) with respect to the size of the corpus.**

The Figure 3 shows the proportion of negative, positive, and neutral tweets concerning the total size of each corpus. Overall, the manual evaluation for all corpora contains more negatively tagged tweets, in contrast to Azure's Text Analitycs tool which tends to tag tweets positively. IBM Watson NLU, on the flip side, labels most tweets as neutral, SAET is more balanced in terms of the proportion of the three labels, but it is observed that it is more effective at labeling tweets with negative polarity. On the other hand, the polarity means were negative for the three corpora, for all tools except for Analytics Text, which obtained an average positive polarity.

Regarding to the metric Recall, the SAET tool obtains greater values in the negative tweets, oppositive to the positives or neutral tweets that were slightly lower. The average Recalls for the three corpora were approximately equal to 60%. In the case of Azure Text Analytics, the Recalls obtained were superior to 60% for positive tweets groups, contrarily to IBM Watson NLU, which achieves high Recalls when rating neutral tweets. This trend was repeated for the three corpora. Similarly, in the case of the metric Precision, the three tools obtain very close results.

When analyzing the Accuracy achieved by the three tools, SAET has higher values in Corpus 1 and Corpus 3, being much higher than the Accuracy achieved by IBM Watson NLU and very close to the results of Azure Text Analytics. This means that in proportion to the number of tweets of each class, SAET is effective in assessing the polarity of sentiment.

As regards the processing and response speed of each tool, both Azure Text Analytics and SAET take less than a second to give the results of the sentiment analysis of each corpus, whereas IBM Watson NLU takes at least four minutes for each corpus. In general, the results show that the performance of SAET is comparable with the performance of commercial tools, presenting an additional advantage, as it is an open-access tool, highly parameterizable.

## 4 CONCLUSIONS AND FUTURE WORK

This work presented the development of a free access web application, SAET, for sentiment analysis on different topics of texts in Spanish within the Ecuadorian context. The results show that SAET has similar performances to compared commercial tools. However, it is more effective for texts with negative polarity. It is important to keep in mind that the manual assessment against which the results of the three tools were compared remains subjective, even though each tweet was weighted independently by six different people. This fact could be the cause of low Precision values. Because SAET performs lexicon-based sentiment analysis, it allows you to easily adjust and customize the polarity lexicon and the sentiment valence shifter dictionary, obtaining results adapted to the contexts.

In future works, to achieve greater precision, it is recommended to adapt the polarity calculation of the *SentimentR* package, incorporating parameters that take into account the grammatical structures of the Spanish language.

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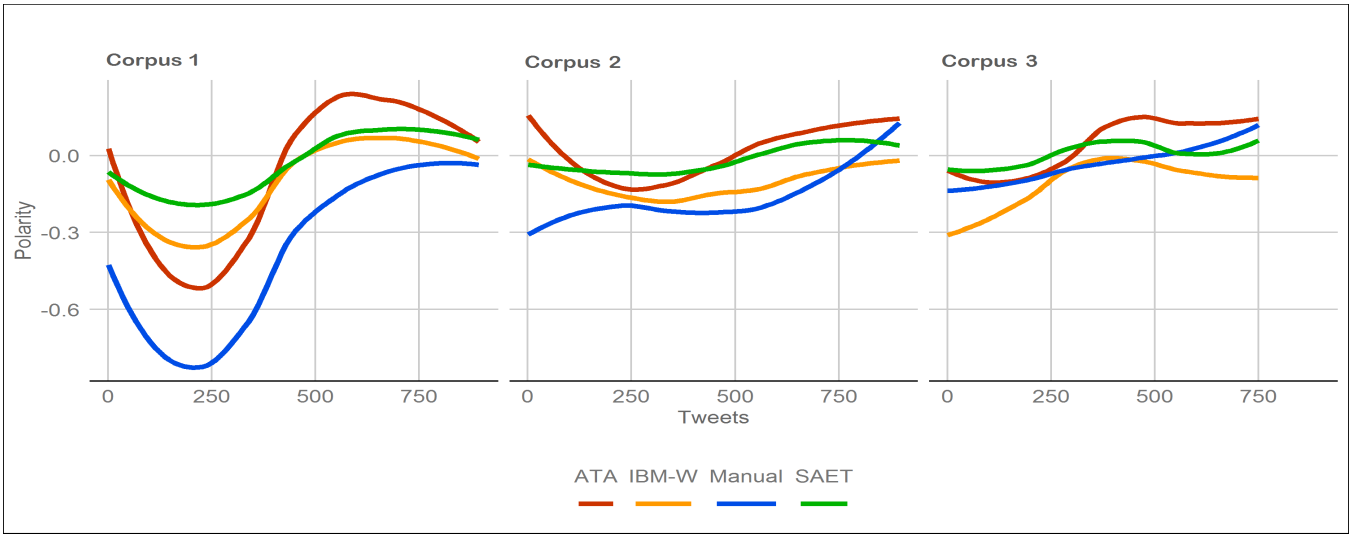



**Figure 4: Manual evaluation trend vs. Azure Text Analytics, IBM Watson, and SAET polarity trends**

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